



**Verified Carbon
Standard**

IMPROVED COOKSTOVES DISTRIBUTION IN RWANDA

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Prepared by	<i>Pozzani Matteo Eni S.p.A. Address: Via Emilia 1, San Donato Milanese (MI), Italy Telephone: Email: matteo.pozzani@eni.com Website: https://www.eni.com/it-IT/home.html</i>

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	4
1.3	Sectoral Scope and Project Type	5
1.4	Project Eligibility	5
1.5	Project Design	8
1.6	Project Proponent	11
1.7	Other Entities Involved in the Project	11
1.8	Ownership.....	12
1.9	Project Start Date	13
1.10	Project Crediting Period	13
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	13
1.12	Description of the Project Activity	14
1.13	Project Location	14
1.14	Conditions Prior to Project Initiation	17
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	20
1.16	Double Counting and Participation under Other GHG Programs	21
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	21
1.18	Sustainable Development Contributions	22
1.19	Additional Information Relevant to the Project	23
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	24
2.1	Stakeholder Engagement and Consultation.....	24
2.2	Risks to Stakeholders and the Environment.....	28
2.3	Respect for Human Rights and Equity	29
2.4	Ecosystem Health	30
3	APPLICATION OF METHODOLOGY.....	32
3.1	Title and Reference of Methodology	32
3.2	Applicability of Methodology	32
3.3	Project Boundary	35
3.4	Baseline Scenario	36

3.5	Additionality	37
3.6	Methodology Deviations	38
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	39
4.1	Baseline Emissions	39
4.2	Project Emissions	39
4.3	Leakage Emissions	39
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	39
5	MONITORING	45
5.1	Data and Parameters Available at Validation	45
5.2	Data and Parameters Monitored.....	48
5.3	Monitoring Plan.....	48
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....	56
	APPENDIX X: <TITLE OF APPENDIX>	ERRORE. IL SEGNA LIBRO NON È DEFINITO.

1 PROJECT DETAILS

1.1 Summary Description of the Project

The clean cooking project developed by Eni S.p.A. referred as “Eni”, is a voluntary Grouped Project (“GP”) involving the distribution of Improved Cookstoves (“ICSs”) among local communities in the Eastern province of Rwanda (“Country”), where most households still rely on inefficient and pollutant traditional cooking techniques.

The activities in field are performed by Eni together with a Project Instance (“PI”) field partner called Rwanda Women’s Network (“RWN”). RWN is a private and non-profit association with the aim of education, solidarity, social promotion, and community development with focus on gender empowerment operating in the Country since 1997, creating a deep and strong relationship with local communities. RWN oversees communities’ engagement, ICSs production, ICSs distribution (storage, management, last-mile distribution and households training) and follow up with recipient beneficiary households for assisting them in the use of the ICS.

The project aims to replace the traditional cooking systems currently used by local people (i.e. three-stone fires or rudimental braziers) with ICSs that require less fuel and thus decrease the biomass combustion’s emissions, with a consequent reduction of health hazards coming from smoke pollution.

The switch to more efficient stoves will help limiting the demand of woody biomass, reducing the unsustainable forests cuttings, and contrasting deforestation which is progressing rapidly in the Country. The improved cookstoves’ deployment will also have a positive impact on women’s and children’s life, reducing the time they spend collecting wood and cooking meals, allowing them to invest their time in other activities.

The ICS beneficiaries voluntarily participate in the GP, signing an End User Agreement (“EUA”) with Eni confirming that any emission reductions resulting from the use of the ICS delivered is owned by Eni. The personal information of the beneficiary household and the serial number of the ICS are noted in the EUA and stored in a data management system which allows Eni to uniquely identify each ICS by tracking the GPS coordinates of the beneficiary household and the ICS delivery date.

According to the project framework, ICSs will be distributed to those households that prior to project implementation, as confirmed during baseline data collection, were relying on the use of three-stone fires or traditional braziers with no improved combustion air supply or chimney, for their cooking activities. Each household targeted by the GP will receive an ICS according to the type of woody biomass currently used as cooking fuel (wood or charcoal).

The current baseline scenario is characterized by the usage of inefficient cooking techniques with poor ventilation and high heat loss. To achieve the same thermal energy, according to the Clean

Cooking Alliance¹, traditional cooking methods could require more than twice the amount of wood fuel compared to the ICS's utilization.

The prosecution of the baseline scenario would see:

- Increasing woody biomass consumption with consequent aggravated deforestation and/or forest degradation and loss of biodiversity in the Country;
- Increasing distances and time to be walked by women and children to collect wood for cooking;
- Increasing incidence of health risks for women and children who inhale the smoke and particulate matter produced by the combustion with traditional systems.

This project will be developed in the Nyagatare district.

The ex-ante estimation of average annual GHG emission reduction generated by the implementation of the first PI of this grouped clean cooking project in the Country, entitled "Project Instance 1" ("PI1"), is around 37,600 tCO₂e per year, for a total of around 0.37 MtCO₂e in the entire 10 years project's lifetime (from 14th December 2023 to 13th December 2033), with lower emission reduction in the initial and last years of the project and a peak in the middle of the project's lifetime due to the ICSs distribution profile.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/verification	Q1 2025	VCS	RINA	One year

1.3 Sectoral Scope and Project Type

Sectoral scope ²	03 – Energy Demand
Project activity type	Type II - Energy Efficiency Improved Project;

1.4 Project Eligibility

¹ <https://cleancooking.org/wp-content/uploads/2023/08/Climate-Environment-and-Clean-Cooking.pdf>

² Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4.1 General eligibility

The project activity is included under the scope of the VCS Program in the category of efficiency improvements in thermal applications. Indeed, the project activity is not mentioned in the list of activities excluded from the VCS Program Scope (Table 1, VCS Standard v4.7)

The methodology applied to the project (Energy efficiency and fuel switch measures in thermal applications, VMR0006 v1.2) is eligible under the VCS Program because it has been developed and approved by Verra.

The VCS Program also excludes the following project activities under the circumstances indicated in Table 1, below, as indicated in the VCS standard version 4.7.

Table 1: Excluded Project Activities

Exclusions from VCS Program Scope	Exclusion
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	Excluded in non-LDCs. Further, large-scale projects excluded in LDCs. Since the PI is meant to deliver improved cookstove to local communities, the technology does not involve any grid-connected electricity generation.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	Excluded in Non-LDCs As mentioned above, the PI is meant to distribute improved cookstove, fueled by wood and charcoal, and does not generate grid – connected electricity
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Excluded in Non-LDCs The PI does not involve heat recovery for combined cycle electricity generation, cogeneration and trigeneration activities

Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO2 capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves)	Excluded in Non-LDCs The end user of the technology provided by PP are the households. Moreover, The stoves are designed for domestic use, not industrial use, so the project and the technology don't generate electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Excluded for large-scale projects in non-LDCs The PI does not involve any fuel switch nor the use of fossil fuel
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year.	Excluded for large-scale projects in non-LDCs The PI does not involve any activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year	Excluded for large-scale projects in non-LDCs The PI does not provide activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Excluded in all countries

	The PI does not involve activities that reduce HFC-23 enusion.
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Since the project does not belong to any categories listed in table above, it can be considered eligible under the VCS program.

1.4.2 AFOLU project eligibility

This section is not applicable because the GP concerns with the energy efficiency improvement of thermal applications adopted by local communities for cooking.

The GP does not belong to the AFOLU projects category.

1.4.3 Transfer project eligibility

This section is not applicable because the GP has been listed solely in the VCS Program and it is seeking registration only in Verra registry.

The GP has not been listed in other carbon registries.

1.5 Project Design

The project has been identified as “Grouped Project”.

1.5.1 Grouped project design

According to the framework of Grouped Projects, each new Project activity Instance is going to meet the eligibility criteria required by VCS Standard v4.7 and the methodology applied (VMR0006 v1.2). The eligible criteria are summarized in the following Table 2.

Table 2: Set of eligibility criteria for new Project activity Instances

No.	Criterion	Explanation of how the new Project activity Instances match the criteria
1	Meet the applicability conditions set out in the methodology applied to the project	The ICSs distributed in a new PI will meet the applicability conditions set out in Section 4 of VMR0006 methodology and Section 2.2 of AMS-II.G methodology: ICSs will be issued to households. ICSs deployed are single pot portable stoves with at least 25% thermal efficiency and powered by woody biomass (wood or charcoal).

		<i>Non-renewable biomass has been used in the regions of new Project activity Instances since 31 December 1989.</i> <i>The double counting of emissions reduction will be avoided collecting the ICSS' serial number and the GPS coordinates of each beneficiary household.</i> <i>ICSSs manufacturers and wholesalers cannot claim for carbon credits generation from project devices due to the purchase agreement signed between the parties.</i> <i>An agreement is signed between the households involved and Eni: each household receives the ICS without cash uptake in exchange for Eni ownership of the environmental benefits' property resulting from the ICS use.</i>
2	<i>Use the technologies or measures specified in the project description.</i>	<i>The ICSSs distributed for new Project Instances will have at least the same technical features of the ones delivered at the start of the project, and in any case higher than 25% as set out in VMR0006 methodology.</i>
3	<i>Apply the technologies or measures in the same manner as specified in the project description.</i>	<i>The ICSSs will substitute the inefficient cooking techniques that are still adopted by the local communities. During the ICS delivery, information regarding biomass use as fuel and cooking baseline appliances will be collected and stored in a data management system to facilitate the following validation and verification processes.</i>
4	<i>Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.</i>	<i>The new Project activity Instances will be issued in the geographical areas identified in Section 1.13, therefore the baseline scenario is the same described in Section 3.4 (traditional inefficient cooking methods with wood or charcoal).</i>
5	<i>Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.</i>	<i>The activity method will be applied for the demonstration of additionality for each new Project Instance.</i> <i>Step 1: Regulatory Surplus</i> <i>The improved cookstoves distribution is not mandated by any government enforced law, policy, statute or other regulatory framework in the Country.</i> <i>Step 2: Positive List</i> <i>The improved cookstoves included in new PI activities will be distributed at zero cost to the households. The only project's source of revenue is the economic value of the carbon credits generated by the introduction of ICSSs.</i>
6	<i>Occur within one of the designated geographic areas</i>	<i>The new PIs activity will be in the geographical area included in the project boundaries defined in the Section</i>

	<i>specified in the project description.</i>	<i>1.13. The GPS coordinates of each beneficiary household will be noted and stored in a management system as evidence of PI implementation inside the project boundaries.</i>
7	<i>Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.</i>	<i>Every new PIs activity will be compliant with the eligibility criteria listed in this table.</i>
8	<i>Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.</i>	<i>Each new PI activity will be described in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria.</i> <i>The evidence of local stakeholders engagement and baseline data surveys and collection inside the project areas are described in this Project Design Document ("PDD").</i> <i>The monitoring data will be stored in a data management system database that will allow the Validation and Verification Body ("VVB") to check the compliance with VCS Standard v4.6 and the methodology VMR0006 v1.2.</i>
9	<i>Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).</i>	<i>The ownership of each new Project activity Instances will be demonstrated by project proponent as the same described in Section 1.8.</i>
10	<i>Have a start date that is the same as or later than the grouped project start date.</i>	<i>The start date would be later then the grouped project start date. The grouped project start date is the 14th of December 2023.</i>
11	<i>Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project,</i>	<i>The project is eligible for crediting since the crediting period and project activity instances start from the date of first ICS distribution, as reported in section 1.9</i> <i>The eligibility of possible additional PI will start from the later of start date of the project activity instance or the start</i>

	through to the end of the total project crediting period	of the verification period in which they were added to the grouped project.
12	Not be or have been enrolled in another VCS project	The grouped project and the correlated project activity instances are not enrolled in another VCS project, neither of them have been enrolled in another VCS project
13	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9	The PI adheres to the clustering requirements for multiple project activity instances. Moreover, the PI adhere to the capacity limit requirements for multiple project activity instances set out in the AMS.II-G methodology, as shown in the sheet <i>Capacity Limit</i> in the emission reduction files

1.6 Project Proponent

Organization name	<i>Eni Natural Energies S.p.A. – Rwanda Branch</i>
Contact person	<i>Martufi Mauro</i>
Title	<i>Branch Manager</i>
Address	<i>Blue Star House, 35 KG 7 Ave, Kigali, Ruanda</i>
Telephone	-
Email	<u>mauro.martufi@eni.com</u>
Organization name	<i>Eni S.p.A.</i>
Contact person	<i>Matteo Pozzani</i>
Title	<i>Carbon Offset Technologies Specialist</i>
Address	<i>Via Emilia 1, San Donato Milanese (MI), 20097, Italy</i>
Telephone	<i>375 590 8939</i>
Email	<i>matteo.pozzani@eni.com</i>

1.7 Other Entities Involved in the Project

Eni has involved Rwanda Women's Network ("RWN"), a Rwandan Non-Governmental Organization, for the field activities implementation. RWN supports Eni with community engagement, ICS purchase, distribution of ICS and follow up and assistance to beneficiary households.

In the agreement signed between Eni and RWN, a clause regarding carbon credits rights has been included stating that Eni is the sole owner of carbon credits resulting from the project activities implementation and that RWN does not have any kind of claim on these.

Organization name	<i>Rwanda Women's Network</i>
Role in the project	<i>Director & Founder</i>
Contact person	<i>Mary Balikungeri</i>
Address	<i>KG 504, Kagugu, Kigali, Rwanda</i>
Telephone	<i>-</i>
Email	<i>rwawnet@rwanda1.rw</i>

1.8 Ownership

Eni Natural Energies S.p.A. - Rwanda branch is the owner of the project, and Eni S.p.a. is the co-proponent of the project (both referred as "Eni")

During the distribution of the ICSs to the households, a member for each household signs an EUA transferring to Eni the property rights of the carbon credits generated with the ICS.

Each ICS is labeled by a unique serial number attached to it that allows to identify every single device. Beneficiary households are informed during ICS training and sensibilization activities that they will receive the ICS free of any cash payment as part of this clean cooking project that Eni is developing in the Country with the goal of generating carbon credits through the distribution and use of improved cooking techniques.

The EUA signed by the recipient households contains the personal information of the beneficiary (i.e. birth date, address, ID document etc.), the house's GPS coordinates and the serial number of the ICS provided. Moreover, Eni's carbon credits ownership is clearly stated in the EUA with the following wording (English text):

"The recipient accepts the assignment by ENE Rwanda of the improved cookstove identified below ("cookstove") without any up-front cash payment undertaking to follow the conditions in this contract....[]

[...]...The recipient waives any rights on the environmental benefit in favour of Eni: the Recipient declares that he/she has understood the mechanism for the carbon credits generation and

waives any claim concerning the ownership of the carbon credits that will be generated using the cookstove”.

1.9 Project Start Date

Project start date	<i>14th December 2023</i>
Justification	<i>The GP started on December 14th 2023 when the first ICS of PI1, was delivered.</i>

1.10 Project Crediting Period

Crediting period	<i>Ten years, fixed</i>
Start and end date of first or fixed crediting period	<i>14th December-2023 to 13th December 2033</i>

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The GHG Emissions Reductions (ERs) have been computed assuming an initial number of ICS planned for the PI1 equal to 40,000, considering the ICS’ parameters declared by producers, such as the expected lifespan.

About the actual operation of ICSs, a conservative assumption has been made for the ERs ex-ante calculation: the ICS that reaches the expected lifetime is considered as non-operating, although some of the devices could still be operational beyond the expected lifetime. In addition, considering the possible loss of operational devices even before the end of their lifespan (i.e. ICS broken or damaged, beneficiary household not using the ICS to its potential etc.), a decreasing ICSs operation rate (“drop off rate”) of 5% per year has been assumed and included in the ER calculation for the 10-year crediting period. If needed, this drop off rate will be updated along the project according to the actual data obtained through ICS usage monitoring.

During the monitoring activities, the ICSs still operating after their expected lifespan may be considered for crediting during the project lifetime; analogously ICSs found out of service before the end of their lifespan will be discounted. The drop off rate parameter will be updated at every monitoring period.

The estimated GHG emission reduction expected at entire project lifetime is 0.37 MtCO₂e with an annual average GHG emission reduction lower than 300,000 tCO₂e/year (project), letting the project falls under “project” category.

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
14-December-2023 to 31-December-2023	139
01-January-2024 to 31-December-2024	18,890
01-January-2025 to 31-December-2025	50,251
01-January-2026 to 31-December-2026	61,979
01-January-2027 to 31-December-2027	57,450
01-January-2028 to 31-December-2028	53,147
01-January-2029 to 31-December-2029	49,059
01-January-2030 to 31-December-2030	45,096
01-January-2031 to 31-December-2031	30,731
01-January-2032 to 31-December-2032	9,542
01-January-2033 to 13-December-2033	0
Total estimated ERRs during the first or fixed crediting period	376,284
Total number of years	10
Average annual ERRs	37,628

1.12 Description of the Project Activity

The project aims to distribute improved cookstoves to households that are still relying on inefficient cooking techniques (i.e three-stone fire or rudimental stove).

The ICSs will reduce GHG emissions from woody biomass combustion thanks to their higher thermal efficiency, with a direct reduction in firewood or charcoal consumption. The ICS design allows to increase the heat transfer to cooking pots by providing a higher thermal insulation of stove's materials and a forced air flow conveyance through the stove mouth.

The switch of cooking systems leads to many co-benefits, first a decrease in woody biomass consumption reducing the pressure on forest cutting in the project area. The reduction of woody biomass necessary for cooking will result in time savings for women and children who are usually involved in biomass collection, allowing them to spend time for other activities.

Moreover, households living in villages far from forests will also save money since they will have to buy smaller quantities of charcoal and wood for cooking.

Furthermore, considering that sometimes the households cook inside their homes, the substitution of traditional cooking mechanisms with ICSs could significantly reduce indoor air pollution with a direct positive impact on the health of household components.

The local communities in the project area are engaged on the framework activities starting from the stakeholder consultations, that gave the population the opportunity to information regarding the project implementation and ask questions/clarifications.

The households receive the ICS free of cash payment. A member of each household involved signs an EAU with Eni waiving the environmental benefits resulting from the use of the device back to Eni.

Every household in the Project Area will be involved in the ICSs distribution if responds to the following requirements:

- current use of woody biomass (firewood or charcoal) as cooking fuel;
- lack of access to cleaner cooking fuels (i.e. gas or electricity);
- lack of efficient cooking device (i.e. intermediate cookstove);
- non-participation in other ICS distribution programs.

The Project Proponent collects the personal data of the designated household's member, taking notes of the house's GPS location and the unique ID number of the delivered ICS. According to the methodology, those data are building up a database of all the beneficiaries which will be the basis for the monitoring and verification activities.

When the household receives the ICS, a dedicated training session is conducted to explain to the family members how the device works and how it is to be used and maintained.

Considering the current cooking fuel utilization (firewood or charcoal), each household involved in the project will receive an ICS powered by the same biomass they currently use as fuel. The project has been launched distributing imported ICS. In the meanwhile, a cookstoves producers scouting is ongoing to identify reliable local producers with the necessary capabilities to fulfil the technical

requirements as per Verra methodology VMR0006 v.1.2. (*“the ICSs distributed for new Project Instances will have at least the same technical features of the ones delivered at the start of the project, and in any case thermal efficiency higher than 25% as set out in VMR0006 methodology”*).

The local production will be implemented gradually to assure that the production meets the targets of the project in terms of ICSs volume and quality, creating more local job opportunities while strengthening local industry in the country.

The two models imported are shown in Figure 1. The two respective datasheets will be attached at the end of this document.



Figure 1: Cookstoves imported in the early phase: wood model (left) and charcoal model (right)

The firewood ICS models have an average lifetime of 7 years, and the charcoal ICS models have an average lifetime of 5 years, as reported by the producer, which means that there will be multiple ICSs deliveries to cover the overall project lifespan (10 years).

According to the technical data sheet provided by the producer, the thermal efficiency for the wood fuelled stove is 50.1%, and the thermal efficiency of the charcoal stove is 44.2%.

In the following table the main technical features of the two cookstoves model imported are presented.

Table 3: Technical specifications of the ICS distributed in the project (manufacturer specifications)

Characteristics	Wood stove	Charcoal stove
Height (mm)	260	275
External diameter (mm)	260	280

Characteristics	Wood stove	Charcoal stove
Weight (kg)	7.5	6.0
Thermal efficiency (%)	50.1	44.2
Lifetime (y)	7	5

According to the baseline scenario, non-renewable woody biomass (firewood or charcoal) is being used by households for their cooking activities with very inefficient appliances (three-stone fire or traditional bracers). Without this clean cooking project implementation, the woody biomass consumption trend would increase with consequent harmful effects on deforestation and loss of biodiversity in the Country.

1.13 Project Location

The project is deployed in Eastern province of Rwanda, considering the districts listed in Table 3. PI1 started in the Nyagatare district.

The Nyagatare City's geographical coordinates are the following:

- 1°18'52.38" S
- 30°22'20.71" E

To manage the different stages of the project, the area of intervention is divided as follow.

Table 3: Project areas

No	Project Province	District
1	Eastern Province of Rwanda	Nyagatare, Gatsibo, Kayanza, Rwamagana, Bugesera, Ngoma, Kirehe

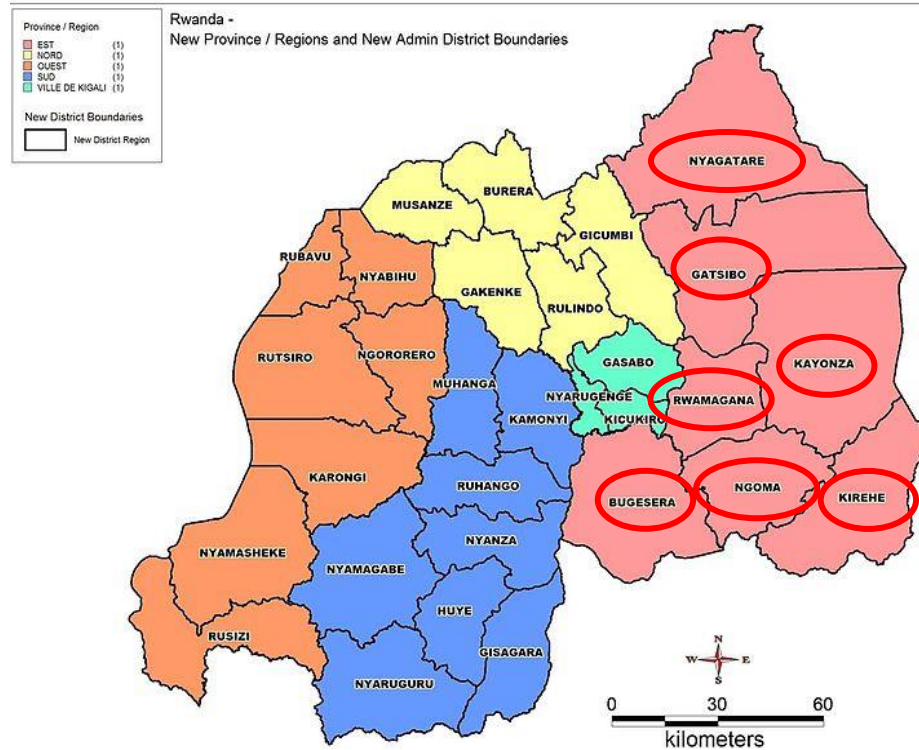


Figure 2: Geographical map of Rwanda with province representation

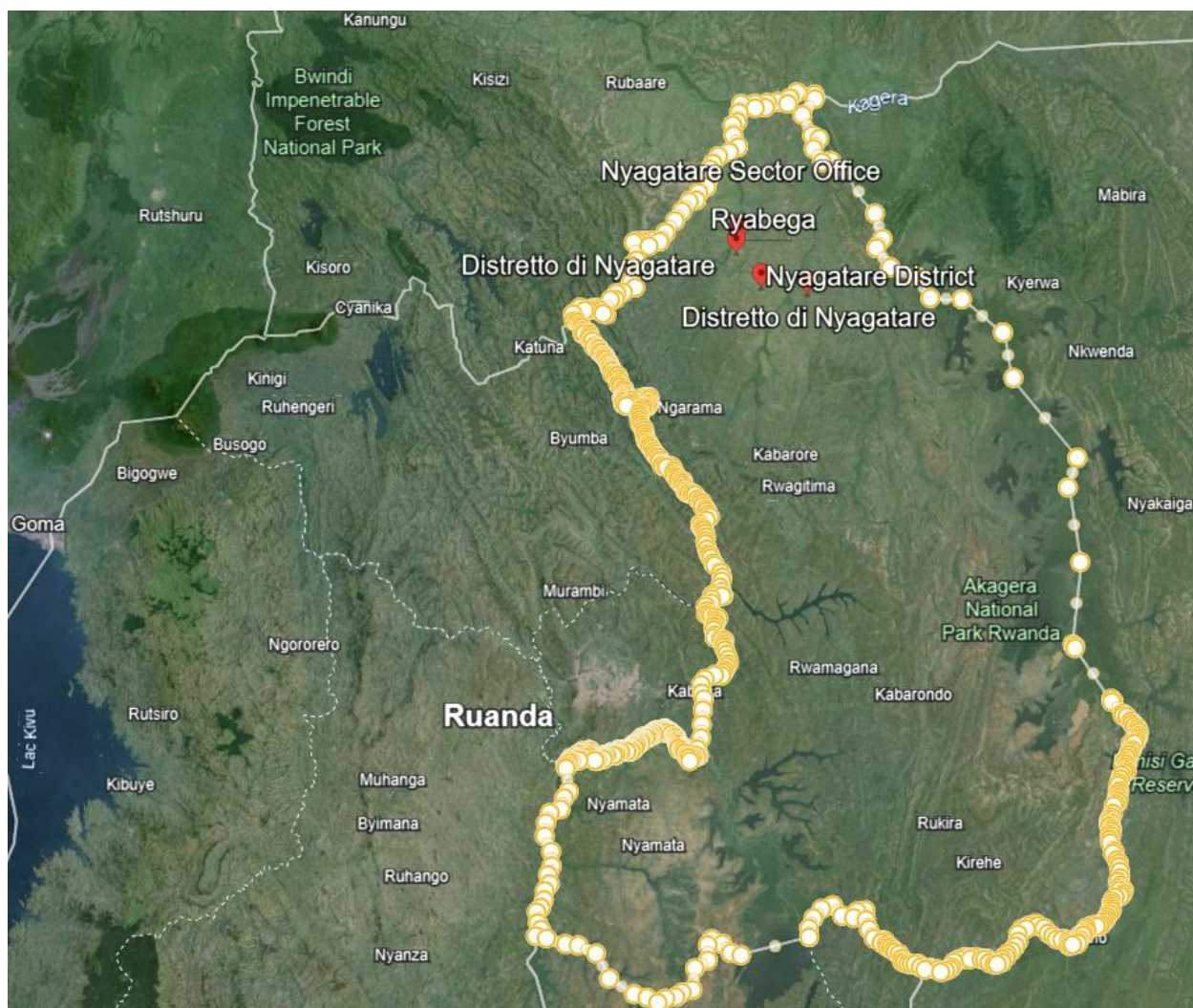


Figure 3: Orthophoto of Rwanda with the Nyagatare district area identification (source: Google Earth)



Figure 4: Orthophoto of Rwanda with the Eastern Province area identification (source: Google Earth)

1.14 Conditions Prior to Project Initiation

The conditions prior to the GP start are equal to the baseline scenario in the overall project area: utilization of inefficient rudimental braziers or three-stone fire burning non-renewable woody biomass (firewood or charcoal). Please refer to Section 3.4 for more details.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

In the Country there is not any mandatory law, policy or regulation which would oblige households to use energy efficient cookstoves. The Country has established a Testing Laboratory for ICSs, providing a certification under the Rwanda Standards Board. The ICSs used in this project will be certified under those standards.

This project is also aligned with the Country NDCs: Rwanda plans to distribute ICSs to 80% of the rural population and 50% of the urban population by 2030.

Before to launch the ICS distribution, the project framework has been shared and discussed with Rwandese authorities in charge of clean cooking activities, who agreed on distribution targets in terms of area and local communities.

The literature review on this field includes:

- Rwanda launches the first ever cook stove testing laboratory²
- Update of the First Nationally Determined Contribution to the United Nations Framework
- Convention on Climate Change (Republic of Rwanda, 2020)³
- The Rwanda Energy Policy (REP, 2015)
- Rwanda Energy Sector Strategic Plan (2018/19 - 2023/24)

² REMA website

³ UNFCCC website

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system?

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 0 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The distribution of ICSs is one of the activities that may accelerate the access to “clean cooking”, an urgent development issue with significant benefits for public health, gender equality, deforestation tackling and climate change fight. The specific activities implemented in the project that will lead to Sustainable Development (“SD”) contributions are:

- Reduced use of biomass fuel for cooking activities;
- Reduced time spent for cooking and collecting fuel;
- Reduced smoke and GHG emissions from cooking activities;
- Engagement and training process of local communities on health, safety and environmental issues;

All these outputs are achieved by promoting and facilitating the switch to more efficient cooking technologies. The GP will contribute to several sustainable development goals, as explained in the SD VISta project document linked to this project. The main contributions are the ones described below:

- SDG3: Ensure healthy lives and promote well-being for all at all ages

According to UNDP, the use of polluting fuels and technologies in traditional ways of cooking is causing respiratory illness, heart problems and even death in many developing Countries. According to the World Bank, the life expectancy in Rwanda is 66 years in 2021. Indoor air pollution due to the use of inefficient cooking appliances caused more than 24,500 deaths in 2019, mainly from respiratory infections being the sixth risk factor for death and disability in the Country.³

- SDG5: Achieve gender equality and empower all women and girls

The energy sector in Sud-Saharan Countries has been widely studied from a gender perspective. The women have the responsibility for domestic tasks, including fuel provisioning and cooking. The use of improved cookstoves would reduce the current practices that are very time consuming: collecting fuel and cooking. The possibility to save time can engage women in other income generating activities, pursue educational

³ <https://data.worldbank.org/indicator/SH.STA.AIRP.P5?locations=RW>

opportunities or simply rest. Moreover, the impact is also related to the women exposure to assault during the collection trips.

- SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all

Clean cooking activities are crucial to reach the SDG7. According to the World Bank database, in 2020 the population' share with access to modern energy cooking services in Rwanda was only 5%. This value goes down to 0.7% considering the rural population.⁴In the report "State of Access to Modern Energy Cooking Services" (World Bank, 2020), the main obstacles are a combination of underdeveloped infrastructures, high up-front capital costs and low households' awareness. In the past decades, the interventions to increase energy access have not been sufficient to keep the path with population growth: therefore, the number of households accessing electricity has grown, but also the absolute number of people cooking with traditional stoves that burn biomass inefficiently has increased. The distribution of ICSs with a thermal efficiency more than double compared to the traditional cooking mechanisms allows the local communities to reduce the biomass collection/purchase, with a direct impact on the households' budget.

- SDG13: Take urgent action to combat climate change and its impact.

The less polluting cooking technology and the decreasing rate in deforestation are both actions to deal with climate change impacts. In the Country's Intended Nationally Determined Contributions submitted in May 2020, Rwanda declared its intention to reduce gross GHG emissions by 4.6 Mtons by 2030, equivalent to an estimated reduction by 38% compared to business as usual (BAU) scenario.⁵

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

According to the AMS-II.G (recalled in VMR0006 v.1.2 methodology), a gross-to-net adjustment factor of 95% has been applied to the savings to account for leakages.

1.19.2 Commercially Sensitive Information

No information has been excluded from the public version of the Project Description.

1.19.3 Further Information

All necessary information regarding the GP implementation has been included in the public version of the Project Description Document.

⁴ <https://data.worldbank.org/indicator/EG.CFT.ACCS.ZS?locations=RW>

⁵ (<https://www.environment.gov.rw/news-detail/rwanda-announces-ambitious-climate-action-plan>)

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	<i>GP's stakeholders have been identified as directly involved by the activities implementation (beneficiary households) or side affected by the activities due to their role in the communities and Country (government, ministries, prefects, mayors and NGOs).</i>
Legal or customary tenure/access rights	<i>The distribution of improved cookstove does not require any kind of customary tenure/access rights since the Project Activity is based only on issue the cookstoves to local communities and visit them for periodical monitoring activities.</i>
Stakeholder diversity and changes over time	<i>Among the beneficiary households, no clear differences on social, economic and cultural aspects have been detected, while the stakeholders indirectly affected by the projects show different social and economic level due to their purpose (i.e. Ministry of environment and NGO).</i>
Expected changes in well-being	<i>The most relevant expected improvement on stakeholders well-being is related to the better off health condition of people usually cooking (less smoke and combustion pollution) and increasing of money and time saving (less fuel to be purchased or collected).</i>
Location of stakeholders	<i>The locations of stakeholders are included in the geographic boundaries presented in Section 1.13.</i>
Location of resources	<i>Territories and resources owned by stakeholders are not impacted by the GP.</i>

2.1.2 Stakeholder Consultation and Ongoing Communication

Eni has conducted several local stakeholder consultations with RWN to adequately inform the local communities in the area on the main pillars of GP implementation. The consultations also enabled the stakeholders to express their point of view on the project and ask questions and clarifications; all this information has been integrated in the project design for improvements.

The stakeholder consultations have been carried on December 5th 2023, in Nyagatare.

The dates and locations of the public meetings held are:

- Nyagatare, 5/12/2023, Moon's Charity Garden

The stakeholders invited to take part in public consultations are the following:

- EU Ambassador in Rwanda
- Minister of Infrastructure
- Minister of State
- Permanent Secretary
- CEO of RDB (Rwanda Development Board)
- DG of REMA
- Mayor of Nyagatare District
- Project development partner representative
- International NGOs representative
- Local NGOs representative

For the invitations, different approaches were adopted to ensure that each stakeholder group was adequately reached and informed on the meetings time, location, and subjects.

Ministries and local authorities received a letter with a formal invitation to attend the meeting, with attached a description of the project.

People interested in project activities on the ground, such as NGO members, women associations, social workers and local households have been made aware of the meetings through posters attached in the sector offices of the Nyagatare district close to the meeting location at least two weeks before the day of each consultation. The content of posters was like to the description shared with the political authorities, explaining the project's goals and implementing guidelines, such as:

- A brief introduction of Eni and its decarbonization strategy, describing in simple terms the nature of carbon credits and their role to achieve the decarbonization.
- The main features of the project (lifetime, areas involved, ICS product, type of beneficiary households involved, advantages coming from ICSs adoption instead of relying on inefficient techniques), describing in detail the distribution mechanism and the project additionality: the ICSs will be deployed without any upfront cash payment requested to

the local beneficiaries, signing an agreement with Eni where the household waives any rights on the carbon credits ownership coming from the emissions reduction obtained from the use of the ICS as opposed to the current cooking techniques. The type of ICS distributed was also presented with pictures.

All these points were thoroughly presented and discussed in detail during each consultation, giving the possibility to the stakeholders to ask clarifications or expose their grievances/opinions/suggestions. To maximize the feedback received from the population and allow everyone the possibility to make comments or express concerns on the project a dedicated email address was created, and contact information was shared during the consultations (info.cookstove.rwanda@eni.com).

Before the conclusion of the meeting, an attendance sheet was circulated and signed by the attendees (those with writing ability and willing to share their information), including personal data for their identification (email address, phone number and role/title if relevant).

Additional stakeholder consultations will be held by Eni and RWN when the distribution will reach new districts in the Eastern Province of Rwanda. These additional meetings will be documented in the monitoring reports.

Stakeholder meeting in Nyagatare

Recorded attendance: 274 people

Date of stakeholder consultation	05-December-2023
Stakeholder engagement process	<i>The stakeholders have been engaged via mail or with public announcement and posters in the area around the meeting location.</i>
Consultation outcome	<i>The main outcomes of the consultation have been the ICS distribution mechanism, the project additionality (ICSs deployed without any upfront cash payment), the monitoring activities and the positive impacts on local communities and environment</i>
Ongoing communication	<i>Stakeholders could communicate through a dedicated mailbox (info.cookstove.rwanda@eni.com)</i>
Stakeholder input	<i>Pastor Jacque for the Restoration Church at Nyagatare, appreciated the advantages of improving cooking stoves project. That will help his church members to be in the fellowship on time.</i> <i>Q: Will the stoves be used from anywhere?</i> <i>A: The stoves can be used inside or outside the house, the important thing is to put it on a plain e stable surface.</i>

	Mr. Kazayire from Burumba, noted that once he receives the stove, he will be helping his wife to cook this time around and gender disparity is going to be addressed in Nyagatare. Mr. Bahati Paul from Nyagatare town. This project is going to enable us to cut less trees because we shall be using less firewood. Mary Claire from CNF: Appreciated the expected impact of the use of improved cooking stoves that will promote gender-equality. She noted that women will have enough time to concentrate on other income generation activities due to the reduction on time spent in the kitchen. Q: is the stove smaller than his cooking saucepan? A: This kind of stove has ability to serve for any saucepans families use, but it cannot be used with commercial pans. Mukamusoni Jane from Barija hailed the provision of improved cooking stoves underlining that her eyes were affected by smoke because of using old traditional cooking techniques.
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Figure 4. Stakeholder consultation in Nyagatare City, 5th December 2023

2.1.3 Free Prior and Informed Consent

Obtaining consent	The households' participation at the GP is voluntary and completely free: during the stakeholders' meetings, it has been deeply underlined that there is no enforcement or obligation in GP
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	participation. Before delivering a cookstove to an interested and eligible household (relying on inefficient woody biomass cooking techniques in baseline condition), an End User Agreement is signed by a component of the family declaring that he/she is voluntary taking part in the project. The project does not interface in any case with Indigenous People and Local Communities' rights.
Outcome of FPIC	The stakeholders' meetings' outputs shared in the section above clearly demonstrate that they fully understand the voluntariness and freedom in taking part in the project. Indeed, no clarifications have been requested on the household's engagement procedure

2.1.4 Grievance Redress Procedure

Development process	The stakeholders could communicate any grievances according to their preferences: via email using the GP' dedicated mailbox (info.cookstove.rwanda@eni.com) or talking with the focal points appointed in each village of cookstoves distribution.
Grievance redress procedure	The grievances would be noted on registry at central level and addressed with the stakeholder involved through email or site visit according to the level of reporting.

2.1.5 Public Comments

This section is not applicable at the moment of project registration

2.2 Risks to Stakeholders and the Environment

	Risks identified	mitigation or preventative measure taken
Risks to stakeholder participation	No risk identified	The participation is voluntary and free, no household is forced to get involved in the project. Moreover, different communications channel have been put in place for guaranteeing and facilitate any stakeholders

		comments or grievance on the project.
Working conditions	No risk identified	The suppliers and partners of Eni must strictly respect the rules and guidelines on work labour condition. The HSE requirements are included in each contract in place and represent a cornerstone of Eni activity.
Safety of women and girls	Burning during cooking activities	Even if the cookstoves drastically increase the safety condition of people involved in cooking activities (in general women and girls), cookstove could get burnt. At the moment of cookstoves delivery, households get a training on proper utilization and maintenance to reduce risks of burnings.
Safety of minority and marginalized groups, including children	No risk identified	The cookstoves distribution and monitoring do not affect the safety of minority and marginalized groups. Regarding the children safety, the cookstoves reduces the risks compared with three-stone fire.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Waste generation for cookstoves disposal	When the cookstoves are broken, spare parts are available allowing the households to repair their cookstoves avoiding waste production. Moreover, the cookstoves that could not be repaired shall be collected and raw materials could be reused by cookstoves manufacturer for making new models.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Use the table below to demonstrate the project's respect for rights related to work and labor.

Discrimination and sexual harassment	<i>The Project proponent firmly condemn every form of discrimination based on gender, religion, or cultural differences. Every worker involved in the GP (cookstoves distributors and monitors) has his/her rights on working condition fully granted.</i>
Management experience	<i>Project proponent and field partner have relevant experience on community projects implementation in Rwanda.</i>
Gender equity in labor and work	<i>Workers have same opportunities and conditions; any kind of discrimination is strictly forbidden. The wages are set for any role in the project, independently on gender, religion or family status.</i>
Human trafficking, forced labor, and child labor	<i>All employees involved in GP activities have signed a job contract according with Rwandan law. The child-labor is strictly forbidden according with Project proponent's policies.</i>

2.3.2 Human Rights

The GP fully respects the human rights of people involved in the activities, both workers and beneficiary households. No discrimination is allowed and permitted during the activities' implementation.

2.3.3 Indigenous Peoples and Cultural Heritage

This section is not applicable because the project activities (cookstoves distribution and monitoring) does not affect the cultural heritage or any rights of indigenous people.

2.3.4 Property Rights

This section is not applicable because the project type (cookstoves distribution and monitoring) does not affect the property rights of stakeholders.

2.3.5 Benefit Sharing

The GP does not foresee benefit sharing with local communities because the property rights of land or resources is directly affected through the project implementation.

The environmental benefits generated through cookstoves utilization is waived back by households who gain the cookstoves' ownership without any cash upfront payment.

2.4 Ecosystem Health

Identify and summarize any risks to the environment and the steps taken to mitigate them. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column.

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	Waste generation for cookstoves disposal	When the cookstoves are broken, spare parts are available allowing the households to repair their cookstoves avoiding waste production. Moreover, the cookstoves that could not be repaired shall be collected and raw materials could be reused by cookstoves manufacturer for making new models.
Soil degradation and soil erosion	No risk is identified	The GP does not involve soil utilization.
Water consumption and stress	No risk is identified	The GP does not involve soil utilization.
Usage of fertilizers	No risk is identified	The GP does not involve soil utilization.

2.4.1 Rare, Threatened, and Endangered species

This section is not applicable because the project type (cookstoves distribution and monitoring) does not affect and interface with endangered species.

2.4.2 Introduction of species

This section is not applicable because the project type (cookstoves distribution and monitoring) does not consider introducing any kind of plant species.

2.4.3 Ecosystem conversion

This section is not applicable because the project type (cookstoves distribution and monitoring) does not belong to ARR, ALM, WRC or AcoGs project categories.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VMR0006	Energy efficiency and fuel switch measures in thermal applications	1.2
Methodology	AMS-II.G	Energy efficiency measures in thermal applications of non-renewable biomass	13.1
Tool	CDM TOOL 30	Calculation of the fraction non-renewable biomass	4.0
Tool	CDM TOOL 33	Default values for common parameters	2.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VMR0006	Project activities introduce energy efficiency in thermal application that increase thermal efficiency to reduce the consumption of non-renewable biomass	GP sees the distribution of improved cookstoves to households that are currently relying on inefficient cooking techniques (i.e. three-stone fire or traditional braziers). The households maintain the same woody biomass as cooking fuel (wood or charcoal), reducing the necessary amount for thermal efficiency improvement.
VMR0006	Methodology applicable to both 'Projects' and 'Large Projects'	The GP size matches the "Project" definition criteria according to the rules set out in "VCS Standard v4.7". The estimated annual emission reduction for the PI1 is on average around 25,000 CO₂e tons per year.

VMR0006	<i>Project activities shall be implemented in domestic premises, or in community-based kitchens, or small and medium-sized enterprise.</i>	<i>The improved cookstoves included in the GP activities will be distributed to households for domestic and/or small enterprise activities.</i>
AMS II.G	<i>Introduction of single pot or multi pot portable or in-situ cookstoves with rated efficiency of at least 25 per cent.</i>	<i>The ICSs are, at the very beginning, imported from an international producer. RWN oversees distribution and monitoring.</i> <i>The thermal efficiency of these ICSs, according to the manufacturer's specifications and certified test results, is higher than 25%.</i> <i>If during the GP implementation (PI1 or other hypothetical following Project Activities), cookstoves produced by other manufacturers (local producers or international companies) are purchased to cover potential production gaps and ensure reaching the overall target population, the thermal efficiency of these additional improved cookstoves will be higher than the threshold set by the methodology applied.</i>
AMS II.G	<i>Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.</i>	<i>According to the World Bank statistical data, the forest area has been declining since 1990. Between 1990 and 2010 Rwanda has lost a total of 210,000 hectares⁶, and between 2010 and 2023 it lost 1,210 ha of natural forest⁷, an average of 6,400 hectares on yearly basis. Use of wood fuel for cooking purpose could be addressed as one of the main drivers of deforestation in the Country: according to World Bank Data, the local population that in 2017 was still relying on woody biomass (wood and charcoal) was</i>

⁶ <https://data.worldbank.org/indicator/AG.LND.FRST.ZS?locations=RW>

⁷ data.worldbank.org/indicator/SP.RUR.TOTL?locations=GQ

		79,9%⁸, with different proportion for urban and rural areas. All the data presented head to conclude that non-renewable biomass has been used in the country since 31 December 1989.
AMS II.G	Explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	To avoid double counting of emissions reduction resulting from cookstoves distributed within the GP boundaries, each ICS is identified and labelled with a unique serial number allowing for a clear distinction between the devices and ensuring that ICS are counted only once in the monitoring of the emission reductions. The serial number applied on the cookstove is stated in the End User Agreement signed by a designated member for the recipient household, together with all other necessary data (i.e. ID number, telephone number, geographical coordinates of their house) establishing a unique association between the household and the ICS distributed. All those data will be collected and stored in a data management system to facilitate their control and check.
AMS II.G	Explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The field partner signed a contract with a clear statement that the ownership of VCUs arising from the cookstoves distribution and utilization belongs exclusively to the Project proponent. The cookstoves manufacturer has no claim on carbon credits because there is a contract in place with Project Proponent specifically for cookstoves purchase. The households that receive an ICS will sign a declaration with Eni waiving the VCUs' property rights in favour of the Project proponent itself. All beneficiary households'

⁸

<https://documents1.worldbank.org/curated/en/099845106292213741/pdf/P163358084c95e020b5280fba571d072a2.pdf>

data will be collected and stored in a data management system to facilitate their control and check.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass	CO2	Yes	Major source
		CH4	Yes	Major source
		N2O	Yes	Major source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO2	Yes	Major source
		CH4	Yes	Major source
		N2O	Yes	Major source
		Other	No	No other source identified
	Fuel production and transport	CO2	No	Not relevant and/or negligible. ICSs allows to reduce the non-renewable woody biomass consumption (wood or charcoal), no processing of biomass residues or dedicated plantation will be run
		CH4	No	Not relevant and/or negligible. ICSs allows to reduce the non-renewable woody biomass consumption (wood or charcoal), no processing of biomass residues or dedicated plantation will be run
		N2O	No	Not relevant and/or negligible. ICSs allows to reduce the non-renewable woody biomass consumption (wood or charcoal), no processing of biomass residues or dedicated plantation will be run
		Other	No	Not relevant and/or negligible. ICSs allows to reduce the non-renewable woody biomass consumption (wood or charcoal), no processing of biomass

residues or dedicated plantation will be run

The first Project activity Instance has been started in the Nyagatare district, whose geographical coordinates of Nyagatare City are the following:

- 1°18'52.38" S
- 30°22'20.71" E

The GP will lead to a reduction of emissions thanks to the introduction of improved cooking devices that allow a higher heat generation, saving up to more than half the quantity of the woody biomass currently used, as explained in Figure 5.

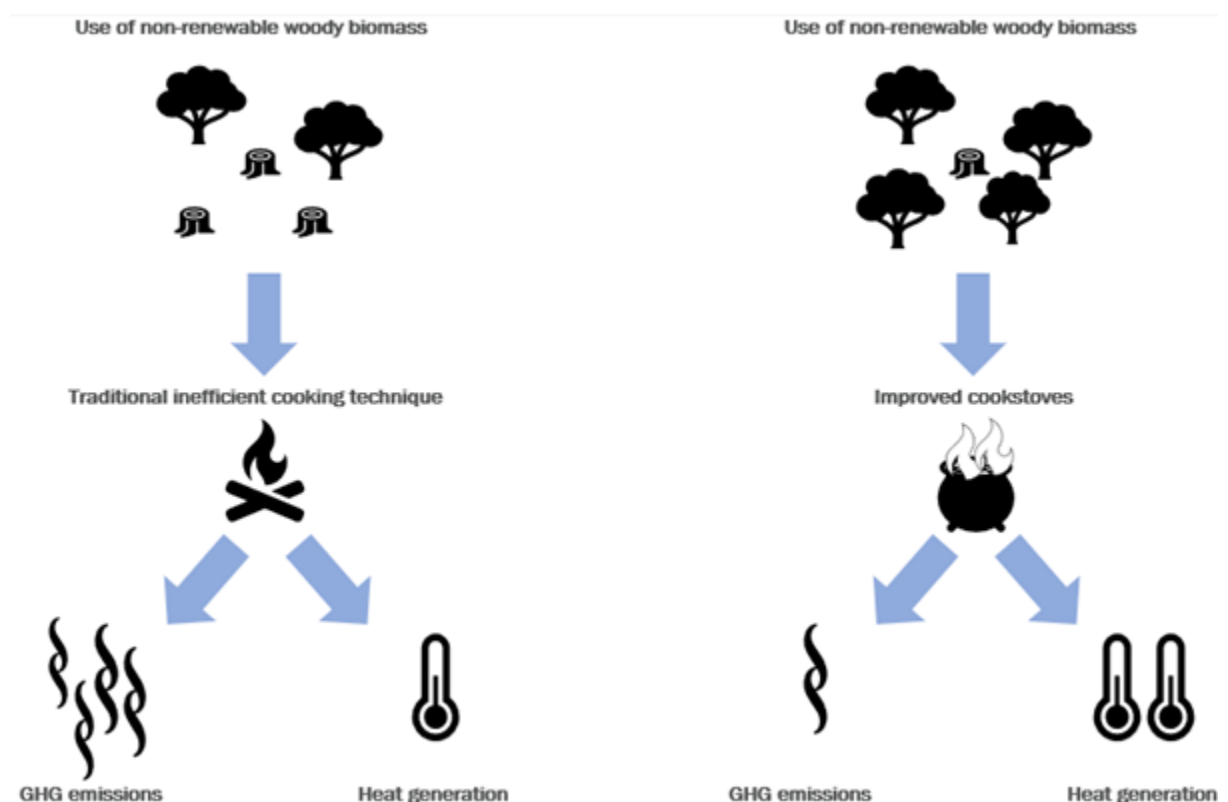


Figure 5: Baseline (left) and Project (right) boundaries for carbon credits generation

3.4 Baseline Scenario

As described in the applied methodology (“VMR0006” v1.2), the baseline scenario would see local communities living in the project area continuing to apply their current cooking habits: use of large amounts of non-renewable woody biomass (firewood or charcoal) as fuel for inefficient cooking methods (three-stone fire or traditional stoves). In the absence of this ICSs distribution,

up to twice the amount of woody biomass would be consumed by the recipient households to generate an equivalent thermal energy for cooking purposes.

The baseline scenario has been defined through the carrying out survey interviews and biomass consumption analyses on the field.

Considering the size of the project area and the overall target population of the project, a random sample of 300 households have been selected and interviewed. The size of the sample used is higher than the minimum sample size for achieving a confidence/precision level of 90/10 as per methodology “VMR0006” v.1.2 request. Since the project will see the distribution of charcoal ICS, the sample included people declaring to use charcoal as main cooking fuel.

The interviews were conducted in the period between 24th and 27th may 2023, in the Nyagatare district located in the same district where the stakeholder consultations have taken place.

In addition to information strictly related to the project implementation (i.e. woody biomass consumption for cooking purpose, cooking techniques/systems used, frequency and duration of cooking activities, willingness to switch to more efficient devices, possible constraints related to this change etc.), the questionnaires also collected other data to gain a deeper knowledge of the local cooking habits (i.e. incidence of health diseases related to smoke inhalation from cooking activities, time spent collecting wood and distances covered, monthly/weekly expenditures for fuel purchase, size of households eating together etc.).

To verify the solidity and reliability of the biomass consumption data gathered through the questionnaires, 150 households have also been involved in the direct measurement of the fuel quantity through the execution of a Kitchen Performance Test (“KPT”). The households involved in the KPTs have been chosen randomly from the interviewed group, trying to gather information on both firewood and charcoal fuel consumers.

The KPTs were performed following the guidelines described in the document provided by the Clean Cooking Alliance (KPT Version 4.0, March 2018), including a qualitative habit survey and daily woody biomass use weighing over four consecutive days.

Since the project will initially distribute exclusively charcoal cookstoves, the KPT targeted only households relying on that specific fuel. Therefore, the results from the baseline surveys and data collection can be summarized as follows:

$H_{\text{woody}} = 2.19 \text{ ton woody biomass/household/year}$

3.5 Additionality

The project activities take place in Rwanda, a Non-Annex 1 country, where the distribution of improved cookstove, and other similar devices, is not mandated by any law, statute or other regulatory framework.

Moreover, the project devices are distributed free of charges for the end users and the project activity has no source of revenue other than carbon credits, and is implemented as part of a government scheme or supported by multilateral funds.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☒ No

3.5.2 Additionality Methods

As per the applied methodology “VMR0006” v1.2, the positive list method is followed for the demonstration of project additionality.

The GP meets all the applicability conditions as described in Section 3.2. Moreover, the GP meets the following conditions:

- The ICSs are issued to beneficiary households without any cash payment from them, hence the GP has no other source of revenue other than the carbon credits generation.
- Project activities are not implemented as part of government schemes, nor the project is supported by multilateral funds.

The GP and each PI fulfil both conditions of positive list, therefore the project is considered as additional.

3.6 Methodology Deviations

This section is not applicable because the project did not apply any methodology deviations.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

This section is not applicable because the methodology does not determine baseline emissions.

4.2 Project Emissions

The methodology “VMR0006” v.1.2 does not account for project emissions separately, rather it quantifies emission reductions as a function of the reduction of non-renewable woody biomass consumption resulting from the sustained use of the ICSs distributed by the GP in substitution of the traditional cooking methods.

4.3 Leakage Emissions

In accordance with Section 8.3 of the applied methodology “VMR0006” v1.2, leakage factor has been assumed as default value equal to 0.95 following the guideline provided in methodology AMS-II.G.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Estimated GHG emission reductions are computed applying Equations 1 of the “VMR0006” v1.2 methodology, for both firewood and charcoal ICSs replacing baseline cooking systems (three-stone fire or traditional braziers):

$$ER_y = \sum_i \sum_j B_{y,savings,i,j} \times N_{0,i,j} \times n_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times (EF_{wf,CO2} + EF_{wf,nonCO2}) \times Adj_{LE} \times (1 - \mu_d)$$

Where:

i = Indices considering more than one type/model of improved cookstove introduced to replace baseline cooking technique (three-stone fire or traditional brazier).

j = Indices considering more than one batch of improved cookstove of type i issued in the project implementation.

ER_y = Emission reductions by improved cookstoves during year y (tCO_{2e}).

$B_{y,savings,i,j}$ = Quantity of woody biomass saved per improved cookstove of type i and batch j (tonnes).

$N_{0,i,j}$ = Number of project devices of type i and batch j commissioned (number)

$n_{y,i,j}$ = Proportion of commissioned project devices of type i and batch j ($N_{0,i,j}$) that remain operating in year y (fraction)

μ_y = Adjustment to account for any continued use of pre-project devices during the year y

f_{NRBy} = Fraction of woody biomass that can be established as non-renewable biomass.

$NCV_{biomass}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default: 0.0156TJ/tonne).

EF_{wf,CO_2} = CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 112 tCO_2/TJ).

$EF_{wf,nonCO_2}$ = Non- CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 9.46 tCO_2/TJ).

Adj_{LE} = Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (fraction).

u_d = Uncertainty deduction for f_{NRB} (%).

The quantity of woody biomass saved by the introduction of the improved cookstoves ($B_{y,savings,i,j}$) is determined by considering the findings of Laboratory test that certified the cookstove. The laboratory applied the Global Appliance Water Boiling Test (WTB) procedure (Version 4.2.3).

- GHG emission reduction estimation ex-ante (preliminary evaluation based on data available at the moment of PD validation phase).

considering the efficiency of the cookstove, the value of fuel saving account to 63.6% for the ICS fuelled by charcoal for the year 1, and 61.5% for the wood ICS. For the following years, an efficiency reduction of 5% is considered (see Table 4 for fuel savings along the cookstove lifetime)

Table 4:

Year	Biomass saving % –ICS charcoal	Biomass saving % –ICS wood
1	66,1%	70,1%
2	64,3%	68,5%
3	62,4%	66,8%
4	60,4%	65,1%
5	58,3%	63,2%
6	-	61,3%
7	-	59,3%

$$B_{y,savings,i,j} = B_{old,i,j} - B_{new,KPT,i,j} = B_{old,i,j} \times Biomass\ saving_{y,i,j}$$

For the ex-ante GHG Emission Reductions:

$$B_{y,savings,i,j} = B_{old,i,j} \times Biomass\ saving_{y,i,j}$$

where:

$B_{old,i,j}$ = Annual quantity of woody biomass that would have been used in the absence of the project activity to generate thermal energy equivalent to that provided by the project device type i and batch j (tonnes/year).

- GHG emission reduction calculation ex-post (identification based on field data coming from monitoring activities such as questionnaires and KPT).

$$B_{y,savings,i,j} = B_{old,i,j} \times B_{new,KPT,i,j}$$

where:

$B_{new,KPT,i,j}$ = Annual quantity of woody biomass used in tonnes per project device of type i and batch j, measured as per the KPT protocol (tonnes/year).

For the GP described, only one improved cookstove will be issued for each household, according to their current cooking fuel (firewood or charcoal). Therefore, the biomass saved will be

associated to the use of just one device per household, with no necessity to consider any woody biomass apportions between multiple improved cookstoves.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
14-December-2023 to 31-December-2023	0	0	0	0	0	139
01-January-2024 to 31-December-2024	0	0	0	0	0	18,890
01-January-2025 to 31-December-2025	0	0	0	0	0	50,251
01-January-2026 to 31-December-2026	0	0	0	0	0	61,979
01-January-2027 to 31-December-2027	0	0	0	0	0	57,450
01-January-2028 to 31-December-2028	0	0	0	0	0	53,147
01-January-2029 to 31-December-2029	0	0	0	0	0	49,059
01-January-2030 to 31-December-2030	0	0	0	0	0	45,096

01-January-2031 to 31-December-2031	0	0	0	0	0	30,731
01-January-2032 to 31-December-2032	0	0	0	0	0	9,542
01-January-2033 to 9-August-2033	0	0	0	0	0	0
Total	0	0	0	0	0	376,284

Ex-ante Parameters	Units	Fixed Project Parameters	Description	Source
E_{f,CO_2}	tCO _{2e} /TJ	112	Wood CO2 emission factor	VMR0006 default
$E_{f,non-CO_2}$	tCO _{2e} /TJ	9.46	Wood non-CO2 emission factor	VMR0006 default
NCV _{biomass}	TJ/tonne of wood	0.0156	Wood net calorific value	VMR0006 default
B _{old,w}	tonnes _w /household/year	2,19	Quantity of woody biomass consumed in baseline (transformation factor from charcoal to wood equal to 4)	Baseline surveys
f _{NRB,y}	Percentage	91.0%	Fraction of non-renewable biomass used in baseline scenario.	CDM methodological tool: "TOOL 30: Calculation of the fraction of non-renewable biomass"

$\eta_{p,c}$	Percentage	44.2%	Efficiency of charcoal project improved cookstove (fraction) at the start of the project activity.	Producer technical data sheet
$\eta_{p,w}$	Percentage	50.1%	Efficiency of wood fueled project improved cookstove (fraction) at the start of the project activity.	Producer technical data sheet
$N_{1,i,j}$	Number	1,000	Number of project devices commissioned in Year 1	Assumption
$N_{2,i,j}$	Number	19,000	Number of project devices commissioned in Year 2	Assumption
$N_{3,i,j}$	Number	20,000	Number of project devices commissioned in Year 3	Assumption
$N_{4,i,j}$	Number	0	Number of project devices commissioned in Year 4	Assumption
$N_{5,i,j}$	Number	0	Number of project devices commissioned in Year 5	Assumption
$N_{6,i,j}$	Number	0	Number of project devices commissioned in Year 6	Assumption
$n_{y,i,j}$	Fraction	0.95	Proportion of commissioned project devices that remain operating	Assumption
Adj_{LE}	Fraction	0.95	Leakage adjustment factor	VMR0006 default
u_d	Percentage	26%	Uncertainty deduction for fNRB,	AMS-II.G default
μ_y	Percentage	100%	Adjustment to account for any continued use of pre-project devices	Assumption

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	<i>B_{old,w}</i>
Data unit	<i>Tonnes/year</i>
Description	<i>Annual quantity of charcoal that would have been used by each household to generate the necessary thermal energy for cooking in the absence of the project activity.</i>
Source of data	<i>Estimated according to options described in CDM methodology “AMS-II.G” v.13, through a field survey as described in Section 3.4 (questionnaires and weighing test, KPT)</i>
Value applied	<i>2.19</i>
Justification of choice of data or description of measurement methods and procedures applied	<i>This parameter has been determined at Grouped Project level; first Project Instance example provided for ex-ante calculations</i>
Purpose of data	<i>Calculation of emission reductions</i>
Comments	<i>Parameter B_{old,c} once determined shall remain fixed for the entire crediting period.</i> <i>A default wood to charcoal conversion factor of 4 kg of firewood per kg of charcoal is considered in line with paragraph 35 of AMS II.G, version 13.</i>

Data / Parameter	<i>eF_{wf,CO2}</i>
Data unit	<i>tCO₂e/TJ</i>
Description	<i>CO₂ emission factor of biomass that is reduced.</i>
Source of data	<i>IPCC default value for firewood</i>
Value applied	<i>112</i>
Justification of choice of data or description of measurement methods and procedures applied	<i>Default value has been assumed according to methodology “VMR0006” v1.2</i>
Purpose of data	<i>Calculation of emission reductions</i>

Comments	-
Data / Parameter	eFwf,non-CO2
Data unit	tCO2e/TJ
Description	Non-CO2 emission factor of biomass that is reduced.
Source of data	IPCC default value for firewood
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Default value has been assumed according to methodology "VMR0006" v1.2
Purpose of Data	Calculation of emission reductions
Comments	-
Data / Parameter	$\eta_{p,c}$
Data unit	Percentage
Description	Charcoal ICS's thermal efficiency at the start of the project activity
Source of data	Manufacturer specification
Value applied	44.2%
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been determined at Grouped Project level; first Project Instance example provided for ex-ante calculations
Purpose of Data	Calculation of emission reductions
Comments	-
Data / Parameter	$\eta_{p,w}$
Data unit	Percentage
Description	Wood ICS's thermal efficiency at the start of the project activity
Source of data	Manufacturer specification
Value applied	50.1%

Justification of choice of data or description of measurement methods and procedures applied	<i>This parameter has been determined at Grouped Project level; first Project Instance example provided for ex-ante calculations</i>
Purpose of Data	<i>Calculation of emission reductions</i>
Comments	-

Data / Parameter	SC
Data unit	Mj / (min x l)
Description	Charcoal ICS's specific fuel consumption
Source of data	Manufacturer specification
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	<i>This parameter has been determined at Grouped Project level; first Project Instance example provided for ex-ante calculations</i>
Purpose of Data	<i>Calculation of emission reductions</i>
Comments	-

Data / Parameter	AdjLE
Data unit	Fraction
Description	Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity
Source of data	CDM methodology AMS-II.G v.13
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	<i>Default value provided by CDM methodology AMS-II.G v.13</i>
Purpose of Data	<i>Calculation of emission reductions</i>
Comments	<i>AdjLE is fixed at validation.</i>

Data / Parameter	ud
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Data unit	%
Description	Uncertainty factor deduction for fnrb
Source of data	CDM methodology AMS-II.G v.13
Value applied	26%
Justification of choice of data or description of measurement methods and procedures applied	According to the CDM methodology AMS-II.G v.13, since the fNRB has been calculated as per CDM TOOL30, ud is defined as default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	Cookstove life span
Data unit	Years
Description	Operating lifetime of ICSs distributed
Source of data	Manufacturers' technical specification
Value applied	5 for charcoal model 7 for wood model
Justification of choice of data or description of measurement methods and procedures applied	Improved cookstoves manufacturer has provided a statement on the expected operational lifetime of the device based on the lab test conducted
Purpose of data	Calculation of emission reductions
Comments	ICSs distribution data will be collected and stored in a Monitoring Database

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project improved cookstoves of type i (firewood or charcoal) and batch j operating during year y
Source of data	Assumption

Description of measurement methods and procedures to be applied	The total number of operational ICS compared to the total number of ICS distributed (according to the monitoring database) will be surveyed. Devices in operation in the sampled group will be counted during each monitoring period to derive an attrition rate (expressed as a percentage) that will be applied to the overall number of ICSs distributed. 90/10 confidence precision will be reached
Frequency of monitoring/recording	At least once per year
Value applied	-
Monitoring equipment	Survey form
QA/QC procedures to be applied	Data from the monitoring survey will be collected at least annually and applied to the emissions reduction calculations (Equation 2) during that monitoring period. Where the required level of precision is not met, additional surveys may be carried out or the lower bound value may be applied.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	Date of commissioning of project device <i>i</i>
Data unit	Date
Description	Actual date of commissioning of the project device
Source of data	Distribution database
Description of measurement methods and procedures to be applied	The distribution date of each cookstoves is recorded and stored in a proper database.
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution
Value applied	-
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	Nd,hh
Data unit	Number
Description	Number of improved cookstoves distributed per household
Source of data	Distribution database
Description of measurement methods and procedures to be applied	The number of improved cookstoves distributed to each household is recorded and stored in a proper database.
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	Each household involved in the GP will get n.1 cookstove. If the cookstove is be damaged or broken, another project device would be given to the households in substitution of the previous cookstoves

Data / Parameter	Bnew,KPT,I,j
Data unit	Tonnes/ICS
Description	Annual quantity of woody biomass used in tonnes per project device of type i
Source of data	Field measures (KPT)
Description of measurement methods	Measured as per the KPT protocol. The KPT shall be carried out in accordance with national standards (if available) or

and procedures to be applied	international standards or guidelines (e.g. the KPT Protocol listed by Clean Cooking Alliance. The days selected for measurement of fuel consumption shall take into account seasonal/weekly variations in fuel consumption, or else the data from the measurement campaign shall be extrapolated in order to take into account the seasonal pattern
Frequency of monitoring/recording	Annually
Value applied	As per CDM methodology AMS-II.G, 90/10 confidence precision will be reached.
Monitoring equipment	Scale
QA/QC procedures to be applied	Data will be collected annually and applied to the emissions reduction calculations during that monitoring period. Where the required level of precision is not met either additional monitoring activities may be completed, or the lower bound value applied.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	μy
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Field surveys
Description of measurement methods and procedures to be applied	Since the baseline devices are most of the time three-stone fire, surveys will be conducted according to CDM methodology AMS-II.G. The surveys shall be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidences to determine the frequency of usage of both the project devices and baseline devices. Field surveys will be performed for each monitoring period, meeting the 90/10 precision requirement

Frequency of monitoring/recording	At least once every two years (biennial)
Value applied	As per CDM methodology AMS-II.G, 90/10 confidence precision will be reached.
Monitoring equipment	Questionnaires
QA/QC procedures to be applied	Data will be collected annually and applied to the emissions reduction calculations during that monitoring period. Where the required level of precision is not met either additional monitoring activities may be completed, or the lower bound value applied.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	ny, i, j
Data unit	Fraction
Description	Proportion of commissioned project devices of type i and batch j (NO, i, j) that remain operating in year y (fraction)
Source of data	Assumption
Description of measurement methods and procedures to be applied	According to CDM methodology AMS-II.G. the project devices still operating shall be measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence/precision levels. Separate samples shall be taken for each batch.
Frequency of monitoring/recording	At least once every year
Value applied	As per CDM methodology AMS-II.G, 90/10 confidence precision will be reached.
Monitoring equipment	Questionnaires
QA/QC procedures to be applied	Data will be collected annually and applied to the emissions reduction calculations during that monitoring period. Where the required level of precision is not met either additional monitoring activities may be completed, or the lower bound value applied.
Purpose of data	Calculation of project emissions

Calculation method	-
Comments	-

5.3 Monitoring Plan

The monitoring plan's goal is to identify solid and reliable estimates of monitored parameter values used in the emissions reduction calculations. The monitoring plan design is based on the CDM Guideline "Sampling and surveys for CDM project activities and programmes of activities" v.9.

Sampling Design

Sampling activities are defined at the Grouped Project level, hence these guidelines will be applied for each Project Instance included in the GP, either individually or in groups.

In accordance with the Standard, the parameters whose confidence precision is not explicitly mentioned in the methodology VMR0006 v 1.2 shall be evaluate with 95/10 confidence precision level since the GP is categorized as "large scale" on CDM base.

In case the required confidence precision level is not met, the lower bound of the confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts.

Parameter values may be estimated by sampling in accordance with Sampling Standard for a single or a group of PI. If required, a single sampling plan covering more PIs will be justified when either the homogeneity of parameters of interest can be demonstrated or the difference among the pls is taken into account in the sample size calculation.

The target population will include ICS recipient households by PI in the monitoring database built up with data coming from data management system created during the iCs distribution through a proper web application able to collect and store all the information required by the methodologies VMR0006 v 1.2 and AMS-II.G. The data monitoring system will be update by field activities partner (i.e. RWN) with information regarding new ICSs distributed (i.e. address, GPS coordinates, ICS ID number, type of fuel etc.), follow-up meetings with beneficiary households and yearly monitoring campaigns (i.e. actual ICS conditions).

For each survey, the sampling method is determined using a random approach on the total ICS beneficiaries chosen from the monitoring database and based on the precision requirements listed above.

The sample size for each monitoring activity will be the total amount of ICS distributed and included in the monitoring database.

Oversampling will be conducted to ensure that the required level of precision is met. The formulas used for sample size definition are the following:

- For proportion-based parameters (i.e. number of operational improved cookstoves):

$$n \geq \frac{1.645^2 \times N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 \times 1.645^2 \times p(1-p)}$$

Where:

n Sample size

N Total number of households involved in clean cooking project

p Expected proportion

1.645 Coefficient that represent the confidence level at 90%

0.1 Coefficient that represent the precision level at 10%

- For mean-based parameters (i.e. woody biomass used with improved cookstoves):

$$n \geq \frac{1.645^2 \times N \times V}{(N-1) \times 0.1^2 + 1.645^2 \times V}$$

Where:

V Standard deviation: $\left(\frac{SD}{mean}\right)^2$

n Sample size

N Total number of households involved in clean cooking project

Mean Expected mean value

SD Expected standard deviation value

1.645 Coefficient that represent the confidence level at 90%

The Sampling Frame will be the monitoring database which may contain information from all pls.

Data Recording

Field-based monitoring activities to determine monitored parameter values could be field survey, KPT or WBT.

Data shall be recorded for each activity either directly onto a smartphone-based app designed for the purpose or, where this is not possible, via paper forms.

Once collected, data will be stored in a monitoring database.

Quality Assurance/Quality Control

To minimize sampling errors, over-sampling will be considered enlarging the sample size in order to ensure a number that meets the required level of precision. This will ensure the integrity of the

sample group and maintains the randomness of participant selection. All sample groups will be re-selected for each monitoring period.

If the required level of precision is not achieved, then further surveys/field activities may be implemented, or lower bound value could be adopted in the calculations with proper justification.

Data analysis

Completed survey questionnaires and KPT/WBT data will be collected by field activities partner (i.e. RWN) and analysed by Project Proponents. After a rigorous cross-check for potential errors and mistakes during the field activities, the monitoring report will be prepared. The monitoring report will include:

- Summary of activities undertaken and main information to explain the context (i.e. location, date, description of activities implemented)
- Summary of data collected and values
- Calculation of the parameter values to be utilized in emissions reduction calculations

Implementation Plan

The implementation of the monitoring plan will be the full responsibility of Project Proponents and their partner as implementing actors (i.e. RWN). The monitoring team will be trained in a proper way to develop all the necessary activities and it will be comprised of local people who speak the local languages and are fully aware of local traditions and habits. Individuals will be selected based on their competence and experience for each monitoring activity and trained accordingly.

To prevent any conflict of interest, there will be no incentives provided for these individuals for any type of data provided

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

There is no sensitive commercial information

Section	Information	Justification